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1 System/Software Requirement Specification (SRS)

1.1 Introduction

1.1.1 Purpose

The primary purpose of SRS document is to define the requirements and expectations for the freelancing web platform, ensuring clarity and alignment among all stakeholders. It serves as a roadmap that guides the development team in creating a product that meets user needs while adhering to the specified constraints and standards. This document also acts as a reference point throughout the development lifecycle, providing detailed descriptions of system functionality, non-functional requirements, and operating conditions. At its core, this document is crucial for addressing the challenges faced by Nepali freelancers in the global market. The purpose of the freelancing web platform is to empower Nepali professionals by offering a localized solution tailored to their unique needs. Throughout this document, we ensure that all aspects of the platform align with the broader objective of fostering a thriving freelancing ecosystem in Nepal.

1.1.2 Intended Audiences and Reading Suggestions

The intended audience for this SRS includes various stakeholders who are involved in or will be affected by the development of the freelancing web platform. The key audiences for this document are:

- **Development Team (Myself):** The SRS will guide me through the project, providing details on the system's features, technical requirements, and structure. It will help me understand the scope and constraints while making development decisions.
- **Project Supervisor:** The project supervisor is a key audience for this document. They will review the SRS to ensure the project aligns with academic standards, meets the required scope, and is feasible within the given timeline. The SRS will help the supervisor assess our understanding of the project and track progress.
- **Tester (Myself):** The SRS will help me define test cases based on the functional and non-functional requirements to ensure the platform works as intended.

1.1.3 Project Scope

The scope of the project is to develop a freelancing web platform aimed at connecting Nepali freelancers with local businesses. The platform will address key challenges faced by Nepali freelancers, such as limited access to international payment systems and competition from global markets, by integrating local payments solutions. The project will focus on providing a localized environment where freelancers can showcase their skills, manage profiles, apply for jobs, and receive payments securely. The platform will also offer a bilingual interface to cater to both Nepali and English-speaking users, ensuring wider accessibility.

1.2 Overall Description

1.2.1 System Perspective

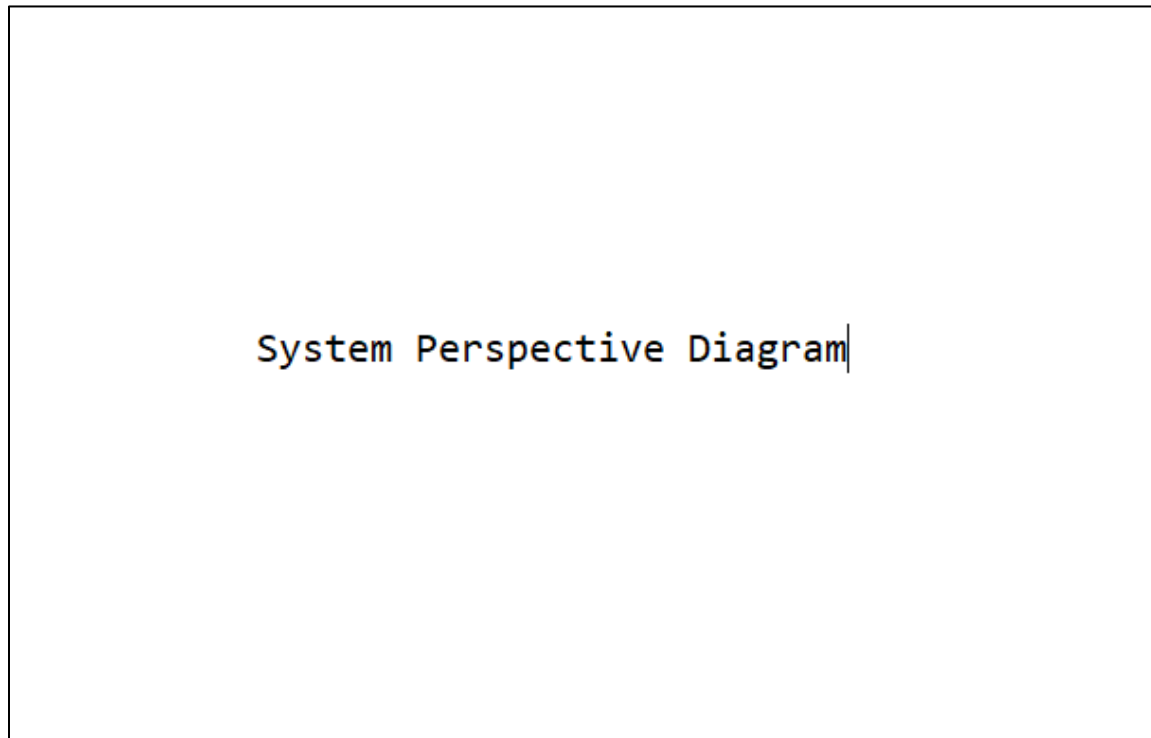


Figure 1: System Perspective Diagram

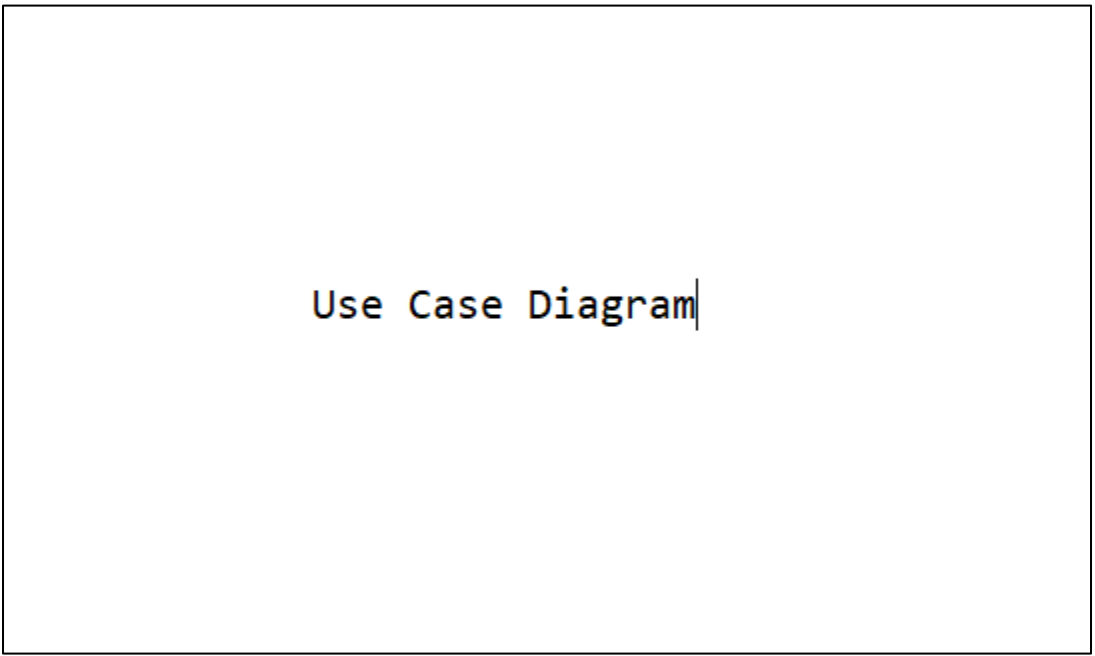
The Application provides an application-based interface to:

SP1: Targeted Vision 1

SP2: Targeted Vision 1

SP3: Targeted Vision 1

SP4: Targeted Vision 1



Use Case Diagram

Figure 2: Use Case Diagram.

1.2.2 System Features

The following are the primary features of the Application.

- **SF1: User Registration and Login**

The platform will allow users (both freelancers and clients) to register and login in using their email or through social media accounts such as Google and GitHub. The registration process will collect basic details like name, email, password, and role (freelancer or client). This feature ensures secure access to the platform and personalized experiences based on user roles.

- **SF2: Job Posting and Management (For Clients)**

Clients will be able to post job listings on the platform, providing details such as job title, description, required skills, budget, and deadline. After posting, clients can manage these listings, view proposals from freelancers, and make updates as necessary.

- **SF3: Profile Management**

Both clients and freelancers will have the ability to create and manage their profiles, which include their skills, experience, portfolio, and hourly rates. They can update their information, showcase completed projects and ensure that their profiles are attractive to potential clients.

- **SF4: Local Payment Integration**

The platform integrates with E-Sewa and Khalti, local payment gateways in Nepal, to facilitate secure and efficient payment transactions between clients and freelancers. This feature allows freelancers to receive payments in local currencies without relying on international banking services.

- **SF5: Bilingual Interface (Nepali and English)**

The platform will support a bilingual interface, allowing users to switch between Nepali and English. This feature is designed to improve accessibility for native Nepali speakers while also catering to non-Nepali speakers, ensuring a user-friendly experience for a broader audience.

- **SF6: AI-Based Project Matching**

The platform will include an AI-based algorithm to match freelancers with relevant projects based on their skills, experience, and past work. The system will analyze user profiles and job listings to suggest the most suitable freelancers for a particular job.

- **SF7: Rating and Review System**

After completing a project, both clients and freelancers will have the ability to rate and review each other based on work quality, professionalism, and communication. This system promotes transparency, builds trust between users, and helps in setting expectations for future engagements.

- **SF8: Bidding System**

Freelancers will have the ability to submit proposals or bids for job postings made by clients. This feature allows freelancers to compete for projects based on their expertise, experience, and pricing. Freelancers can offer customized proposals, and clients can review these bids to choose the most suitable freelancer.

1.2.3 User class and characteristics

The following are the classes of users who will have specific roles of operation with Application and its reports:

UC1: Normal User

- **Description:** A normal user is someone who has not yet registered for the platform. This user class has limited access to the platform, allowing them only to browse general information and explore the features without full engagement. They are considered visitors who can explore the platform before making an account or deciding to take on a specific role.

UC1.1: Sign Up and Role Selection

- **User Role:** Normal users have the option to sign up for the platform by providing necessary details (such as name, email, and password). After registration, they will be prompted to select their role either as a freelancer or a client to unlock additional platform features such as bidding for jobs or posting job listings.

UC1.2: Access to General Information

- **User Role:** The normal user can view general information about the platform, such as the types of services offered, how the platform works, and the benefits of joining as either a client or freelancer. They can explore the homepage, read FAQs, and understand the overall purpose of the platform without creating an account.

UC1.3: Browse Job Listings and Freelancers

- **User Role:** Normal users can browse through job listings posted by clients and view the profiles of freelancers. However, they cannot interact with the job posts (e.g., apply or bid) or send messages to freelancers without first registering.

UC1.4: Search Functionality

- **User Role:** Normal users can use the search feature to look for jobs or freelancers by category, skills, location, and other filters. However, they cannot perform advanced searches or save preferences unless they create an account and select a role.

UC2: Client

- **Description:** Clients are users who post job listings on the platform and hire freelancers for their projects. They are the primary source of job opportunities.

UC2.1: Post Jobs

- **User Role:** Clients can create, post, and manage job listings. They will provide detailed information about the job such as title, description, required skills, budget, and deadline.

UC2.2: View and Evaluate Proposals

- **User Role:** Clients can view proposals and bids submitted by freelancers. They can evaluate the freelancers' profiles, ratings, and pricing to select the most suitable candidate for their job.

UC2.3: Make Payments

- **User Role:** Once the work is completed, clients will make payments through the platform using local payment gateways (E-Sewa or Khalti) to the freelancers. They will also be able to leave ratings and reviews.

UC3: Freelancer

- **Description:** Freelancers are users who offer their services on the platform. They can browse job listings posted by clients, submit proposals or bids, and engage in direct communication with clients

UC3.1: Browse and Search Job Listings

- **User Role:** Freelancers can browse, and search job listings posted by clients based on various categories, skills, and locations. They can filter and find suitable jobs that match their expertise.

UC3.2: Submit Proposals and Bids

- **User Role:** Freelancers can submit proposals or bids for job listings. They provide a detailed proposal outlining their skills, work experience, and price for completing the job.

UC3.3: Communicate with Clients

- **User Role:** Freelancers can send messages to clients to clarify job requirements, negotiate prices, or discuss project details. Communication happens through the platform's messaging system.

UC4: Admin/Developer

- **Description:** The Admin/Developer is responsible for overseeing the management and maintenance of the platform. They ensure that the platform is secure, operational, and that all features are functioning as expected.

UC4.1: User Management

- **User Role:** The admin can view, manage, and delete user accounts (both clients and freelancers) in case if the account is inactive.

UC4.2: Manage Job Postings

- **User Role:** The admin can monitor all job postings on the platform. They can approve, edit, or delete job listings to ensure that all posts comply with platform guidelines and are appropriate.

UC4.3: Platform Settings and Configuration

- **User Role:** The admin can configure system settings, such as payment gateways, language options (Nepali/English), and other platform configurations that affect the general operation of the system.

UC4.4: Platform Maintenance

- **User Role:** Admins will handle platform updates, bug fixes, and performance monitoring to ensure smooth operation.

1.2.4 Operating Environment

- **OE1:** The application is designed to operate primarily on a web platform, accessible through modern web browsers like Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge. The current version of the platform is intended for desktop users. In the future, support for mobile platforms, including Android and iOS, may be considered through responsive design or native applications.
- **OE2:** The platform is designed to function seamlessly on all modern operating systems, including Windows, macOS, and Linux for users accessing it through a web browser.

Operating System and Platform Picture|

1.2.5 Design and Implementation Constraints

- **CO1: No mobile app version**

Explanation: The application will be a web-based platform and will not have a dedicated mobile app version. Although a mobile app could enhance user accessibility, this feature is outside the scope of the current project, which focuses on web functionality.

Solution: The platform will be fully responsive and optimized for mobile browsing. Users will be able to access the platform from their mobile devices through a web browser, but there will be no separate mobile app developed for the project.

- **CO2: Limited payment gateway integration**

Explanation: The application will integrate only the two specified payment gateways (E-Sewa and Khalti) for processing transactions. This is due to the project's scope, which focuses on supporting local payment methods used within Nepal.

Solution: Only local payment systems, E-Sewa and Khalti, will be integrated to handle payments. Clients and freelancers will be able to complete transactions using these two options.

- **CO3: Limited Role Selection during Registration**

Explanation: During the registration process, users will only be able to select one of two roles, Client or Freelancer. There will be no support for users to switch roles after registration or have multiple roles simultaneously.

Solution: The registration form will prompt users to select either the "Client" or "Freelancer" role, and this selection will define their access and functionality on the platform. Role-switching will not be allowed once the user has completed registration.

1.2.6 Assumptions and Dependencies

- **AS1:** The platform will be optimized for web browsers only.
- **AS2:** The users can use the application for multiple periods of time.
- **AS3:** The platform will primarily be available in the English language, with Nepali as an option.
- **AS4:** Users will need to verify their email during registration.

1.3 Functional Requirements

1.3.1 User Registration and Role Selection

Req.ID	Requirement Description	Priority	Complexity	
FR.01	Users should be able to sign up using their email and password. After registration, they will receive an OTP sent to their email for verification. Once verified, they will choose their role (Freelancer or Client).		Most	High
	System Requirement			
	SR.01	On opening the application, users will be directed to the home page where they can initiate the sign-up process.		
	SR.02	Upon submitting the sign-up form with email and password, an OTP will be sent to the user’s registered email address for verification.		
	SR.03	Users must enter the OTP within a limited time (1 minute) to verify their email.		
	SR.04	After OTP verification, users will be prompted to choose their role (Freelancer or Client) before proceeding to access the platform.		

1.3.2 User Login

Req.ID	Requirement Description	Priority	Complexity
FR.02	Users (Freelancers and Clients) should be able to log in to the platform using their registered email and password.	Most	High
	System Requirement		
	SR.01 Users will be directed to a login page where they can enter their registered email and password.		
	SR.02 If the user enters valid credentials, they will be successfully logged in and redirected to the dashboard.		
	SR.03 If the user enters incorrect credentials, they will receive an error message indicating the issue (e.g., "Invalid email or password").		
	SR.04 The platform will have a "Forgot Password" feature that allows users to reset their password through their registered email.		

1.3.3 Job Posting (Client)

Req.ID	Requirement Description	Priority	Complexity
FR.03	Clients should be able to create and post job listings, providing details like job title, description, required skills, budget, and deadline.	Most	Medium
	System Requirement		
	SR.01 Clients will be able to access a job posting form after logging in.		
	SR.02 The job posting form will include fields for title, description, skills, budget, and deadline.		
	SR.03 Upon submission, the job posting will be saved and displayed in the job listings.		
	SR.04 The platform will notify clients if any required fields are missing.		

1.3.4 Browse Job Listings (Freelancer)

Req.ID	Requirement Description		Priority	Complexity
FR.04	Freelancers should be able to browse job listings posted by clients.		Most	Medium
	System Requirement			
	SR.01	Freelancers will have access to a job listing page where all posted jobs are displayed.		
	SR.02	Freelancers will be able to filter job listings based on categories, skills, and budget.		
	SR.03	Job listings will include basic details like job title, description, and deadline.		
	SR.04	Freelancers will have the option to apply or bid on jobs after registration and role selection.		

1.3.5 Bidding System (Freelancer)

Req.ID	Requirement Description	Priority	Complexity
FR.05	Freelancers should be able to place bids on job listings, providing their proposed price and a short cover letter.	Most	High
	System Requirement		
	SR.01 Freelancers will see a "Place Bid" button on job listings after logging in.		
	SR.02 Freelancers can enter their bid amount and a brief message to the client.		
	SR.03 Once the bid is submitted, it will appear in the job post and be visible to the client.		
	SR.04 Freelancers can update or withdraw their bids before the client selects a freelancer.		

1.3.6 Database Connection

The application requires a reliable and secure database connection to ensure proper storage, retrieval, and management of user data, job listings, transactions, and other platform-related information.

Req.ID	Requirement Description		Priority	Complexity
FR.06	The system will ensure secure and consistent connection with the database for all transactions and operations.		Most	High
	System Requirement			
	SR.01	The application will establish a database connection at server initialization, ensuring minimal latency during user operations.		
	SR.02	All data transactions, such as user registration, job postings, and payments, will be securely handled using encrypted database connections.		
	SR.03	The database structure will be optimized for efficient querying and minimal downtime, supporting scalability for increased usage.		
	SR.04	No manual user access to the database is provided, ensuring security and data integrity.		

1.4 External Interfaces Requirements

1.4.1 User Interfaces.

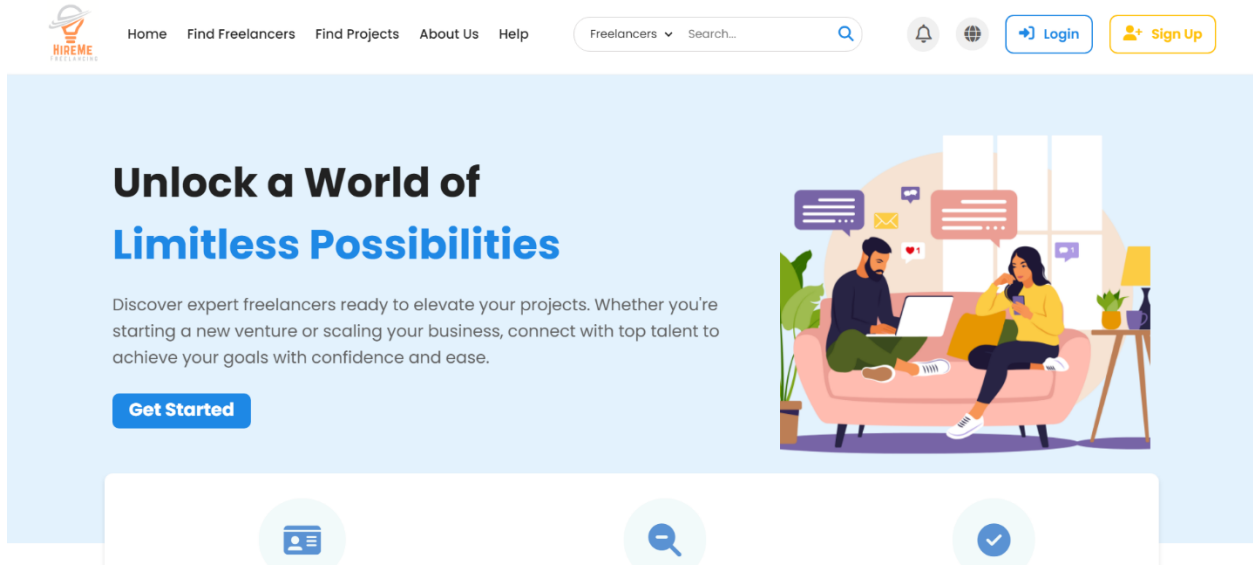


Figure 3: Home Page Interface

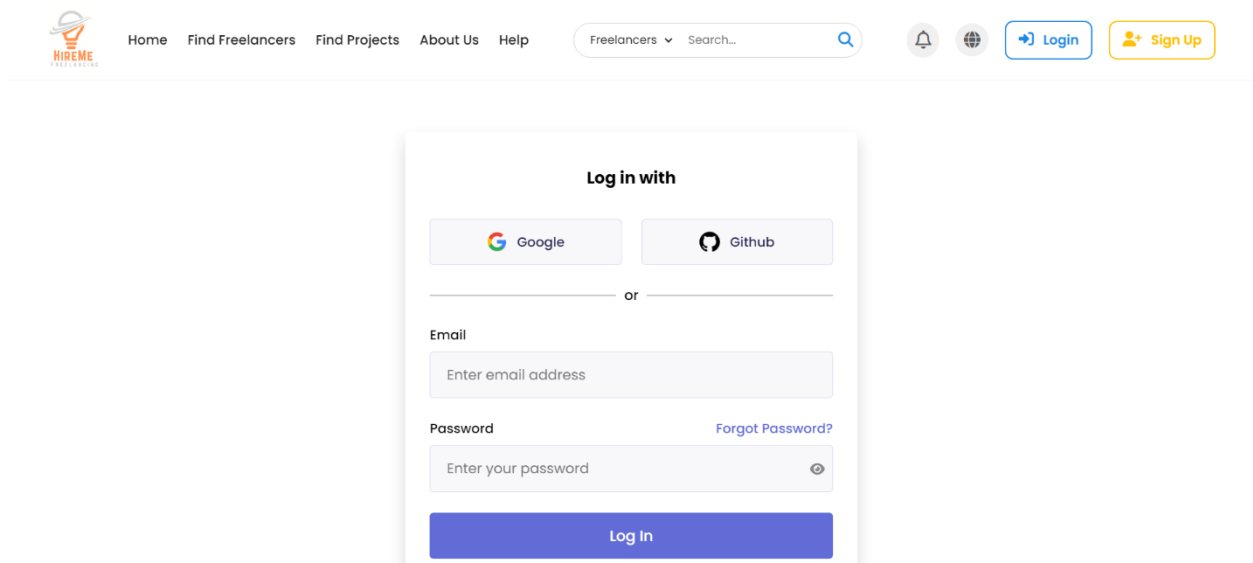
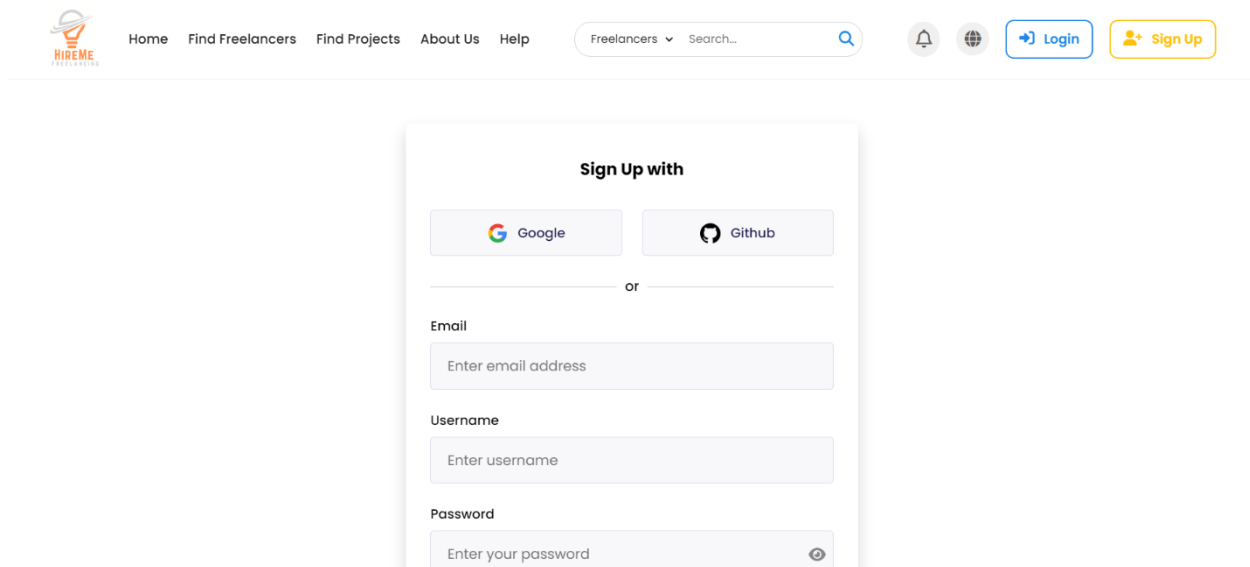


Figure 4: Login Page Interface



The Sign Up page features a top navigation bar with the HireMe logo, links for Home, Find Freelancers, Find Projects, About Us, and Help, a search bar, and buttons for Login and Sign Up. The main content area contains a 'Sign Up with' section with Google and Github login options. Below these is a form with fields for Email, Username, and Password, each with a placeholder text.

Sign Up with

Google Github

or

Email

Enter email address

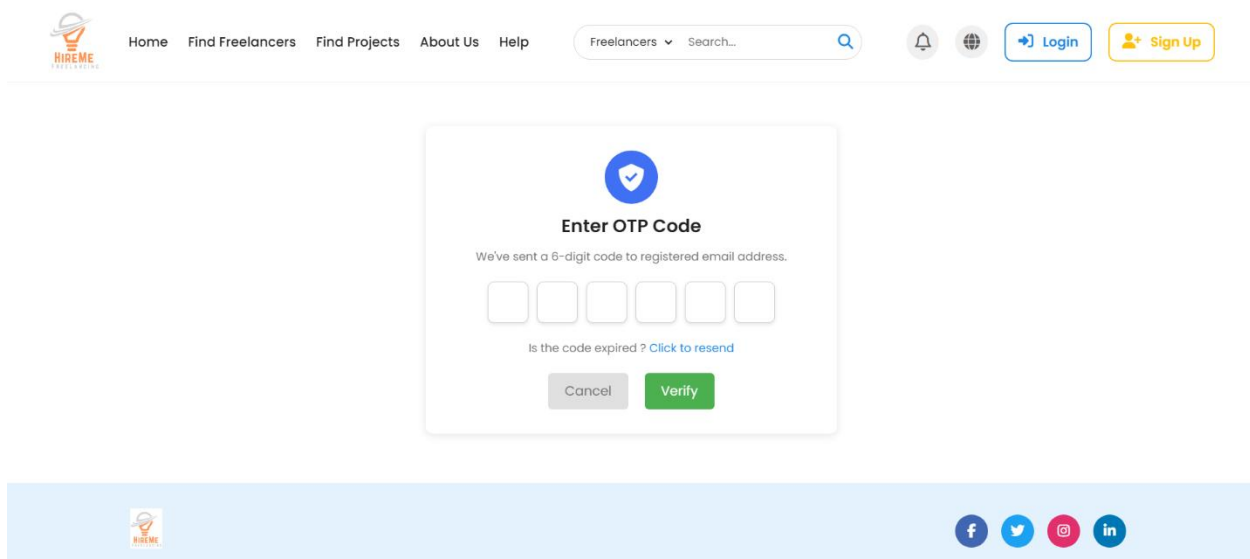
Username

Enter username

Password

Enter your password

Figure 5: SignUp Page Interface



The OTP-Verification page features a top navigation bar identical to the Sign Up page. The main content area contains a 'Enter OTP Code' section with a blue shield icon. Below the icon is a message stating 'We've sent a 6-digit code to registered email address.' followed by six input boxes for the code. A link to 'Click to resend' is provided if the code has expired. At the bottom are 'Cancel' and 'Verify' buttons. The footer contains the HireMe logo and social media icons for Facebook, Twitter, Instagram, and LinkedIn.

Enter OTP Code

We've sent a 6-digit code to registered email address.

Is the code expired ? [Click to resend](#)

Cancel Verify

Figure 6: OTP-Verification Page Interface

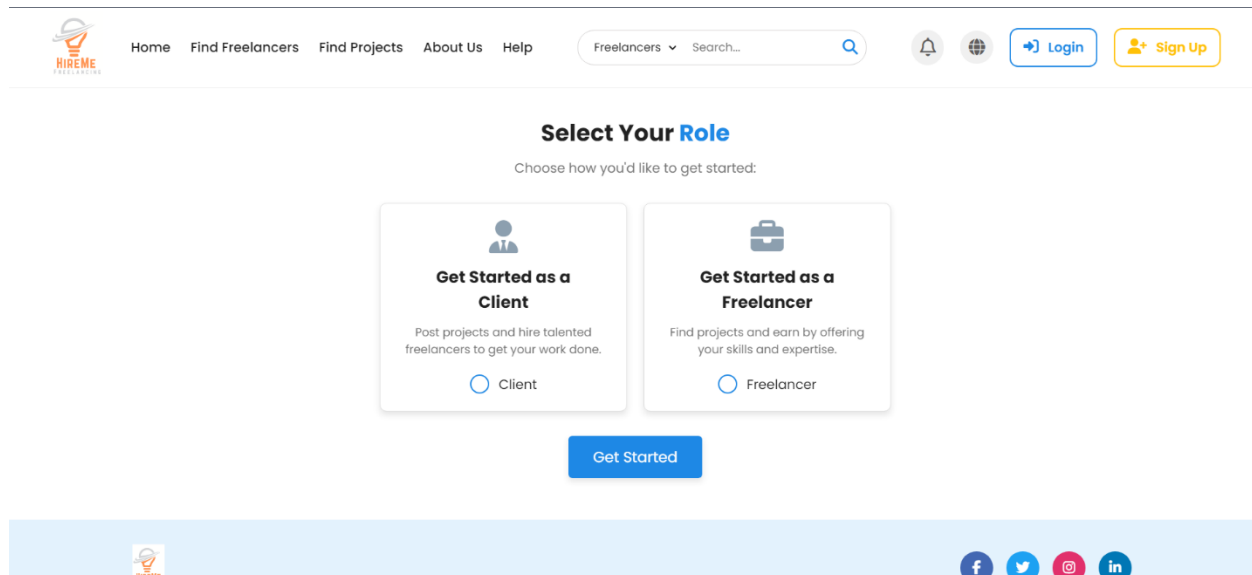


Figure 7: Role Selection Page Interface

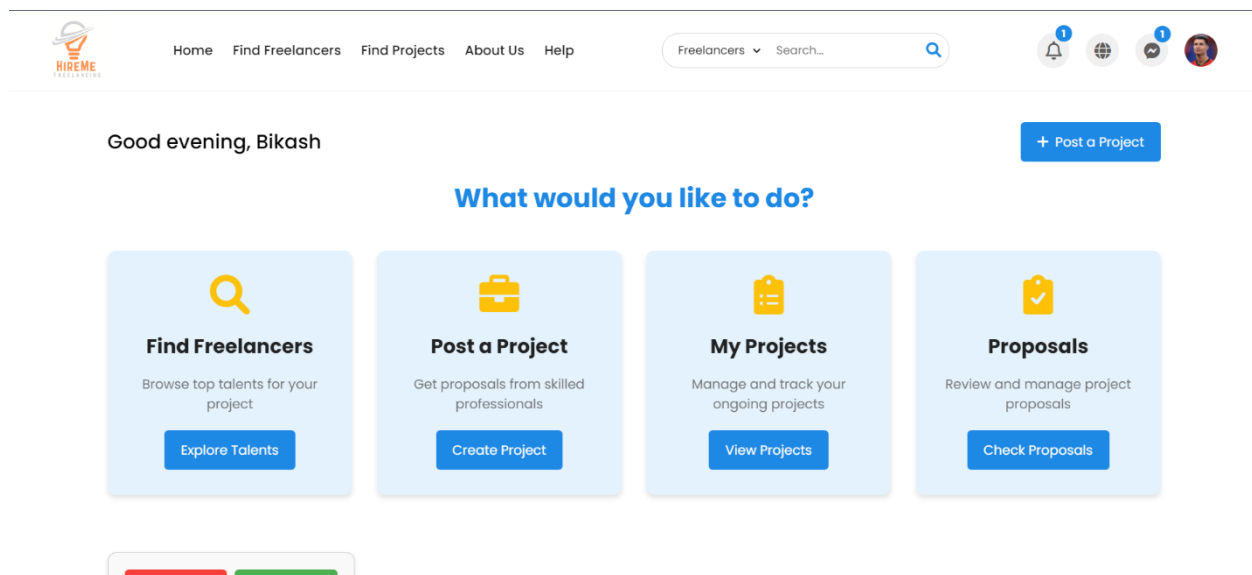


Figure 8: Dashboard Page Interface

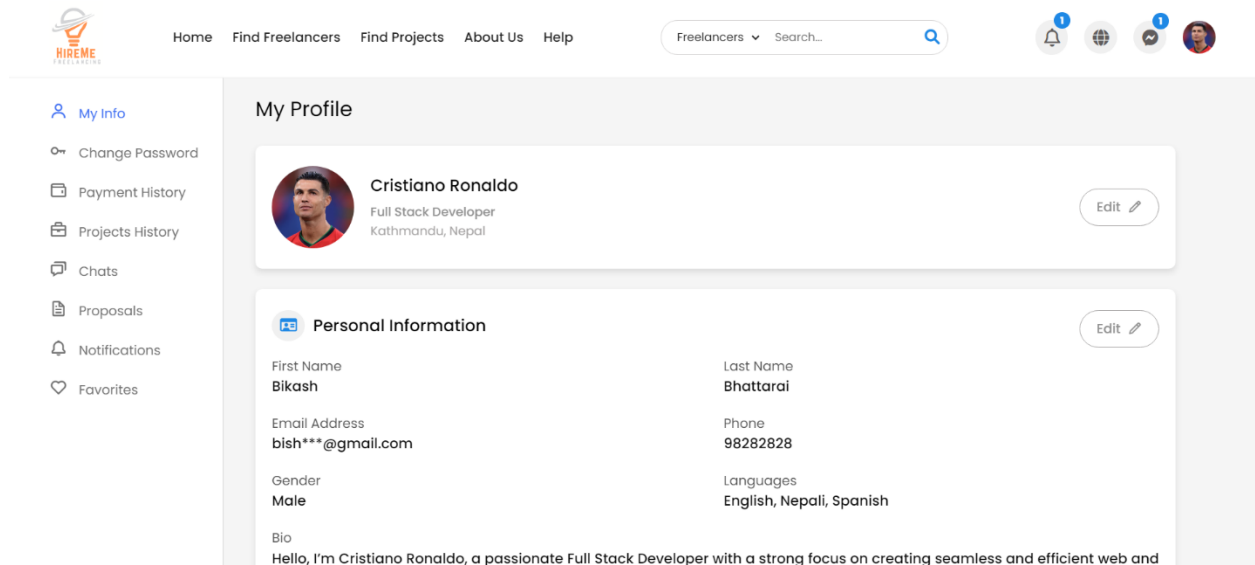


Figure 9: User Profile Page Interface

1.4.2 Hardware Interfaces

The freelancing web platform requires specific hardware interfaces to ensure efficient functionality. On the client side, users need devices with at least 2 GB of RAM and a modern web browser, such as Google Chrome, Mozilla Firefox, or Safari, to access the platform. A stable internet connection is also essential for real-time interactions and data exchanges with the server. On the server side, the application requires a hosting environment equipped with at least 8 GB of RAM, 100 GB of storage, and a quad-core processor to manage backend operations and database interactions effectively. Additionally, the server must have a reliable network infrastructure to handle multiple concurrent users without performance degradation.

1.4.3 Software Interfaces

The application will run on Django, leveraging its built-in features for handling backend operations and rendering frontend templates. For the database, MySQL is used to store user data, job details, and transaction records. End users will interact with the application through modern web browsers like Google Chrome, Mozilla Firefox, or Safari. Local payment gateway APIs for E-Sewa and Khalti will be integrated, allowing seamless transaction processing. The software stack should also include Python (for Django), a MySQL connector, and necessary libraries to support template rendering and form handling.

1.4.4 Communication Interfaces

The application facilitates communication between the server and the end user via standard HTTP/HTTPS protocols. Email notifications for OTP verification and user communications will be enabled using Django's built-in email framework, configured with SMTP services. Payment gateway APIs will use secure RESTful communication to process transactions.

1.4.5 Other Non-Functional Requirements

The non-functional requirements outline the key attributes of the application beyond its functional capabilities. These requirements focus on ensuring that the system is user-friendly, secure, reliable, and maintainable, aligning with the goals of creating a responsive and efficient freelancing platform.

1.4.5.1 Performance Requirements Description

The performance requirements ensure the system operates smoothly under expected workloads and provides a reliable user experience.

Req.ID	Requirement Description	Priority	Complexity
PR.01	Upon opening the application, it should not crash.	Should	High
PR.02	The application should display the homepage within 3 seconds for most users.	Must	Medium
PR.03	The platform should handle up to 100 concurrent users without degradation in performance.	Must	High
PR.04	The search functionality should return results within 2 seconds.	Could	Medium

1.4.5.2 Safety and Security Requirements

Safety requirements ensure that the application operates securely and protects user data.

Safety Requirements

Req.ID	Requirement Description	Priority	Complexity
SR.01	Permission should be asked by the application to the user for sensitive operations.	Should	High
SR.02	The application should not install any malware or spyware in the background.	Should	Medium
SR.03	The system should log unauthorized access attempts and notify the admin.	Could	High

1.4.5.3 Other Software Quality Attributes

Req.ID	Requirement Description	Priority	Complexity
SQA.01	The application should behave consistently across platforms and different versions.	Could	Medium
SQA.02	The application should be clean, user-friendly, and easy to operate.	Should	High
SQA.03	The application should have minimal downtime during maintenance and updates.	Should	High
SQA.04	The platform should use adaptive layouts to support both desktop and mobile devices.	Could	Normal